

Preliminary Design Review & Prototype

Multi-Author Writing Style Change Detection

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What is Author Style Change Detection?

- Detecting points in a document where the **writing style changes**.
- Implies the presence of **multiple authors**.
- Applications:
 - Plagiarism and contract cheating detection.
 - Forensic linguistics and authorship attribution.
 - Analysis of collaborative writing (e.g., Wikipedia).
- This project develops a **web-based tool** that automates detection using stylometric features and machine learning.



Problem Statement

Challenges

- Subtle stylistic differences between authors.
- Lack of large, annotated multi-author datasets.
- Need for **interpretable** results, not just black-box predictions.



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Goal

Build a **proof-of-concept prototype** that:

- Extracts 31 stylometric features from sentence pairs.
- Trains a classifier to predict author switches.
- Provides confidence scores and explanatory features.
- Validates the feasibility of the processing pipeline and user interaction.



Four-Stage Pipeline

Data → Features → Model → Evaluation

Data	Features	Model	Evaluation
Text upload, sentence segmentation, input validation.	Stylometric feature extraction from each sentence pair.	Classifier training (LR, SVM, k-NN, MLP) and author-switch prediction.	Cross-validation, metrics, model comparison.

- **Separation of concerns:** each package can be modified independently.
- Aligns with the conceptual class diagram (next slide).



Architecture Justification

The four-package decomposition provides:

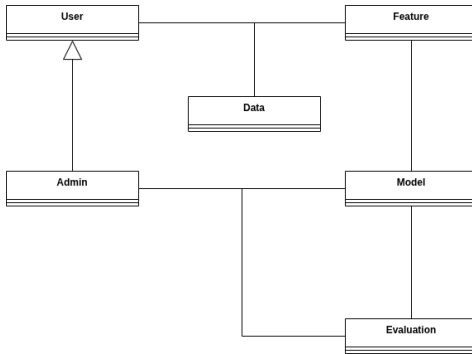
- **Data** – isolates sentence segmentation and input validation changes.
- **Features** – allows new stylometric features without affecting other components.
- **Model** – enables independent replacement or extension of the classification algorithm.
- **Evaluation** – permits separate evolution of metrics and reporting.

Benefits:

- Improved maintainability.
- Simplified unit and integration testing.
- Direct alignment with the domain model (conceptual class diagram).



Conceptual Class Diagram



Core domain entities: **User**, **Admin**, **Data**, **Features**, **Model**, **Evaluation**.

Implementation-specific constructs (sentence buffers, prediction caches) are excluded – deferred to later design phases.



Use Case Traceability

Use Case	Related FRs	Validating Tests	Status
UC1: Upload text for analysis	FR1–FR3	UT-06–11, IT-01–03	Implemented
UC2: View predictions	FR5–FR6	IT-04–06, ST-01–02	Implemented
UC3: Interpret confidence	FR7	ST-01, UAT	Partial
UC4: Train model	FR8	IT-07–08, ST-04	Implemented
UC5: Compare model performance	FR9	IT-11	Partial
UC6: Login and authentication	Auth	UT-01–05	Partial



Unit Test Cases – Authentication (UT-01 to UT-05)

ID	Description	Expected Result
UT-01	Username validation – valid	Username accepted
UT-02	Username validation – invalid	Error message displayed
UT-03	Password validation – correct	Login successful
UT-04	Password validation – incorrect	Login rejected
UT-05	Empty login fields	Validation error displayed



Unit Test Cases – Document Processing (UT-06 to UT-11)

ID	Description	Expected Result
UT-06	Empty document upload	Upload rejected with warning
UT-07	Valid text upload	Document processed successfully
UT-08	Unsupported file format	Upload rejected with error message
UT-09	Oversized document	Upload blocked
UT-10	Sentence segmentation	Sentences split correctly
UT-11	Single-sentence submission	Warning displayed



Unit Test Cases – Feature Extraction (UT-12 to UT-16)

ID	Description	Expected Result
UT-12	Word count extraction	Count calculated correctly
UT-13	Punctuation density	Density calculated correctly
UT-14	Function-word ratio	Ratio computed correctly
UT-15	Type-token ratio	Correct lexical richness value
UT-16	Feature vector generation	Vector generated successfully



Unit Test Cases – Prediction, Training & Evaluation (UT-17 to UT-23)

ID	Description	Expected Result
UT-17	Prediction without model	Informative warning displayed
UT-18	Prediction with trained model	Predictions generated successfully
UT-19	Confidence-score generation	Confidence score displayed
UT-20	Author-switch detection	Switch positions identified
UT-21	Model training execution	Training completes successfully
UT-22	Evaluation metrics	Precision, recall, F1 displayed
UT-23	Model comparison	Comparison table displayed correctly



Integration Test Cases (IT-01 to IT-11)

ID	Description	Expected Result
IT-01	Inspect endpoint – valid text	HTTP 200; sentence cards displayed
IT-02	Inspect endpoint – file upload	HTTP 200; content parsed correctly
IT-03	Inspect endpoint – empty body	Flash warning displayed
IT-04	Predict endpoint – no model	Message: “No model available”
IT-05	Predict endpoint – with model	HTTP 200; predictions shown
IT-06	Predict endpoint – fewer than 2 sentences	Flash warning displayed
IT-07	Train endpoint – dataset missing	Flash error; log shows dataset error
IT-08	Train endpoint – full pipeline	model.pkl created on disk
IT-09	Explore endpoint – statistics	Statistics correct per difficulty level
IT-10	Explore endpoint – both example types	Both columns displayed correctly
IT-11	Model comparison endpoint	Side-by-side metrics displayed



System Test Cases & UAT

ID	Description	Expected Result
ST-01	End-to-end – two known switches	Both switches detected; top-3 features shown
ST-02	End-to-end – single-author	Zero switches detected
ST-03	Performance – 50 sentences	Processing completes in <10 seconds
ST-04	End-to-end – train then predict	Model mode used; F1 displayed
UAT	Proxy user workflow	Usability rating $\geq 4/5$

These preliminary test cases are subject to revision as the system evolves. All 38 entries are documented in the PDR document.



Validation Strategy

Testing Pyramid

1 Unit Testing

- Validates feature extraction and document processing modules independently.
- Target coverage: $\geq 80\%$

2 Integration Testing

- Verifies Flask routes and workflow interactions.
- Uses Flask test client for deterministic testing.

3 System Testing

- Evaluates complete end-to-end user scenarios.
- Includes performance and prediction validation.

4 User Acceptance Testing

- Proxy user performs upload-analyse-review workflow.
- Measures usability and interaction clarity.



Key System Scenarios

Scenario 1: Multi-Author Detection

- User uploads a document containing multiple writing styles.
- System extracts stylometric features between sentence pairs.
- Classifier predicts author-switch positions with confidence scores.

Scenario 2: Single-Author Validation

- Document contains consistent writing style throughout.
- Expected result: zero author switches detected.

Scenario 3: Performance Requirement

- 50-sentence document processed within interactive time.
- Acceptance target: under 10 seconds on reference hardware.



Prototype Overview

- Implemented as a **Flask-based web application** (Python).
- Technologies: Flask, NumPy, scikit-learn, HTML/CSS.
- Pages:
 - **Home** – model status, quick-start links.
 - **Inspect** – upload text, view per-sentence features.
 - **Explore** – dataset statistics and examples.
 - **Train** – classifier selection, training with live log.
 - **Predict** – submit text, probability threshold, results.
- Runs locally via `python app.py`; `http://localhost:5000`.



Functional Requirement Coverage

FR	Requirement	Prototype	Final
FR1	Upload or paste text	Implemented	Implemented
FR2	Sentence segmentation	Implemented	Implemented
FR3	Input validation and errors	Implemented	Implemented
FR4	Stylometric feature extraction	Implemented	Implemented
			(extra)
FR5	Pairwise feature differences	Implemented	Implemented
FR6	Author-switch prediction	Implemented	Implemented
FR7	Confidence and top features	Partial	Implemented
FR8	Admin model training	Implemented	Implemented
FR9	Model comparison	Partial	Implemented
Auth	User authentication	Partial	Implemented

Partial items are implemented with limited functionality in the prototype and will be fully realised in the final system.



Live Demo

Prototype Walkthrough

The prototype will be demonstrated live, covering the Inspect, Train, and Predict workflows.

(Focusing mainly on Data Processing)



Test Plan – Objectives & Levels

Main objectives

- 1 Validate all functional requirements against specifications.
- 2 Classifier macro-averaged F1 > simple baseline on PAN 2026 held-out set.
- 3 Process 50-sentence documents within interactive time (target: < 10 s).
- 4 Display top stylometric features associated with predicted switches.
- 5 Graceful rejection of invalid/empty inputs.
- 6 Correct authentication validation.



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Test levels (pyramid)

Unit $\geq 80\%$ statement coverage; pytest with mocks.

Integration Flask test client – upload, train, predict workflows.

System End-to-end scenarios (multi-author, single-author, empty input).



Test Strategy

■ Approach:

- White-box: unit and integration tests (internal interfaces).
- Black-box: system and acceptance tests (HTTP/UI interfaces).

■ Framework: `pytest` + `unittest.mock`; Flask test client for fast, deterministic HTTP simulation.

■ Test Data:

- Unit tests: hand-crafted sentences with known feature values.
- Integration / System: synthetic dataset with ground-truth labels.
- Model evaluation: PAN 2026 held-out test split.



FR–Test Case Traceability

FR	Requirement	Covering Test Cases
FR1	Upload or paste text	UT-06–09, IT-01–02, ST-01
FR2	Sentence segmentation	UT-10–11
FR3	Input validation	UT-06, UT-08–09, UT-11, IT-03
FR4	Feature extraction	UT-12–16
FR5	Pairwise differences	UT-16
FR6	Author-switch prediction	UT-17–20, IT-04–06, ST-01–02
FR7	Confidence and explanation	ST-01, UAT
FR8	Model training	UT-21, IT-07–08, ST-04
FR9	Model comparison	UT-23, IT-11
Auth	Login authentication	UT-01–05



Test Environment

Component	Details
Laptop Model	Acer Aspire A315-58
Operating System	Fedora Linux 44 (Workstation Edition)
Processor	11th Gen Intel Core i3-1115G4
Memory	8 GB RAM
Python Version	3.10+
Framework	Flask
ML Library	scikit-learn
Testing Framework	pytest
Browsers	Chrome 124+, Firefox 125+



Risks and Contingency

Risk	Likelihood	Mitigation
PAN dataset unavailable	Medium	Use synthetic dataset as drop-in replacement
Classifier performance below baseline	Medium	Ablation study; add features if needed
Time constraints limit UAT participants	High	One proxy instructor sufficient; structured walkthrough
Training too slow for CI pipeline	Low	Mock classifier in integration tests; nightly full training



Processing a Document in a Short Time

Performance Target

The system shall process documents containing up to **50 sentences** within a time frame suitable for interactive use.

Specific goal (from acceptance criteria):

- A fifty-sentence document must be processed in **under ten seconds** on reference hardware.

This requirement is validated by system test ST-03 and ensures the tool remains practical for real-time interactive analysis.



Acceptance Criteria

The system is considered ready for the next development phase when:

- 1 All unit tests pass, with $\geq 80\%$ coverage in core processing modules.
- 2 All documented integration scenarios execute without runtime errors.
- 3 Preliminary classifier outperforms a simple baseline on the PAN 2026 held-out test split.
- 4 A fifty-sentence document is processed in < 10 **seconds** (performance requirement).
- 5 UAT proxy user completes the upload-analyse-review workflow without assistance and gives a usability rating $\geq 4/5$ on a Likert scale.



Prototype Implementation Details

- **Inspect page (FR1–FR4):** text upload, sentence segmentation, 31 stylometric features per sentence, overview statistics, colour-coded style-distance bar.
- **Explore page:** dataset statistics and example display.
- **Train page (FR8):** classifier selection (Logistic Regression, SVM, k-NN, MLP), cross-validation, synchronous training with live log streaming.
- **Predict page (FR5–FR7):** text submission, probability threshold slider, ML-based predictions when a model exists.
- **Results page:** held-out F1, precision, recall, cross-validation statistics.
- **Authentication (partial):** login form and session management; full validation deferred.

Known Limitations:

- Random-baseline predictor when no model is trained (UI exploration only).
- Synchronous training – will be moved to background task queue.



Conclusion & Future Work

Key Achievements

- Design review completed: modular architecture, conceptual class diagram, detailed test plan.
- Working proof-of-concept prototype validates the stylometric pipeline and user workflow.
- All core use cases (UC1–UC6) covered; FRs partially or fully implemented.



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- Design review completed: modular architecture, conceptual class diagram, detailed test plan.
- Working proof-of-concept prototype validates the stylometric pipeline and user workflow.
- All core use cases (UC1–UC6) covered; FRs partially or fully implemented.

Future Work (Final Implementation)

- Asynchronous model training (background task queue).
- Support for PDF/DOCX uploads.
- Full user authentication and model comparison.
- Permutation-based feature importance ranking.



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Thank You

Questions?

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